WEEK 2 · DAY 1 · INSTRUCTOR NOTES

Take It Apart: Trend, Season and Noise

For the trainer only.

The one idea for today: *every series is several simpler stories added together. Pull them apart, and the leftover (the residual) is where the surprises are.*

At a glance

Length	~2 hours
Participants leave able to	decompose a series with `seasonal_decompose`, choose the right period from the process, choose additive vs multiplicative, read a residual, know when to reach for `STL(robust=True)`
Covers (original)	Week 3 Code Book: *Seasonal periods at a glance*, **A1 bus power**, **A2 battery health**, **B2 Ka-band monthly**, **B4 NOC tickets daily**, **C1 additive vs multiplicative**; Week 3 quiz topic *decomposing an operational time series*
Datasets	`ka_band_availability_monthly`, `noc_tickets_daily`, `additive_multiplicative_demo`, `bus_power_telemetry_hourly`, `battery_health_daily` (**week-03 version**)
Files you need open	animated slides, your **solution** notebook, the rule book PDF

The day before: checklist

☐ Run your **solution** notebook top to bottom: all 13 checks should be ✅ (about 30 seconds; STL takes a few seconds).

☐ Re-read the 6 quiz questions for this topic in `Time-series/MCQ/week-03-mcq.csv`. Every one is covered by a slide (see the table below).

☐ Practise the "Lagos traffic" hook out loud. It's the anchor for the whole day.

Run of show (2 hours)

Time	Block	What happens	Slides
0:00	**Warm-up**	Week 1 in 3 lines; quick quiz from the room	1–3
0:08	**Hook**	Lagos traffic = trend + season + noise; the jollof analogy	4
0:12	**The equation**	observed = trend + seasonal + residual, on real Ka-band data	5

Time	Block	What happens	Slides
0:18	**Part 1: the tool**	`seasonal_decompose` recipe, the two errors, reading the 4 panels	6–8
0:28	**Choosing the period**	24 / 7 / 12 from the process; the weekend in numbers; wrong period = more noise	9–10
0:35	**Part 2: add or multiply**	the fare analogy; the log trick	11–13
0:42	**Part 3: the residual**	the August fault; eclipse needs STL; one disaster August; three cycles; battery has two seasons	14–19
0:55	**Hands-on**	Part A (25 min), Part B (35 min). Walk the room.	20
1:55	**Wrap-up**	Recap, cold-call the periods, tease autocorrelation	21

Explain it like this (in plain English)

- **Decomposition** = splitting one line into three: a slow **trend**, a repeating **seasonal** pattern, and the **residual** leftovers. Additive means you can add them back together and get the original exactly (they prove it in A4).
- **The trend** comes from a moving average one period wide. That's why it's empty for half a period at each end (6 months for monthly data). That's normal; say so before someone asks.
- **The period** is the most important setting and the most common mistake. It's not "12 because seasons are yearly". It's *how many rows until the pattern repeats*: 24 for hourly data with a daily eclipse, 7 for daily data with a working week, 12 for monthly data with a yearly rainy season. Use the wrong period and the real pattern gets dumped into the noise (they see the noise nearly double).
- **Additive vs multiplicative:** look at the swing. Same size as the level rises → additive. Grows with the level → multiplicative (or take the log). Fare analogy: +₦100 every December vs +20% every December.
- **The residual** is the investigation list. For an incident review you look at the residual, not the raw series, because the raw series is full of "normal" patterns.
- **STL** is a smarter decomposition that lets the season change slowly. `robust=True` stops one disaster from spoiling the seasonal pattern.

Slide talking points

#	Say this	Ask this
4	"You feel the total. Decomposition separates the causes again."	"Why was your trip long today? Trend, season or noise?"
5	"Add the bottom three lines and you get the top line back. Exactly. They'll prove it in the notebook."	–

#	Say this	Ask this
8	Trend runs **Jul 2023 → Jun 2025**: missing 6 months at each end. August's seasonal effect is **−1.10 points**. Biggest residual only **0.16**: a clean series.	"Why is the trend shorter than the data?"
9	"The period is a fact about the process."	"Daily NOC tickets: which period?" (quiz)
10	Sunday **−2.90** tickets, Saturday **−2.39**, Tuesday **+1.35**. Noise left: **1.23** with period 7 vs **2.27** with period 12.	"Why did the noise nearly double?"
12	Additive swing **0.29 → 0.28**; multiplicative **0.29 → 0.54**.	"Which one is a 20% fare increase?" (quiz)
13	After `np.log` the swing is **0.77 → 0.83**: nearly constant.	–
15	August residual: **18 Aug 09:00** is **11.6 A** off; the next biggest is only **0.59 A**. It's an injected solar-array dropout. Link to NigComSat-1 (2008).	"Where would an incident review look?" (quiz: the residual)
16	Classic decomposition over all 120 days: **all 46** of the top-46 leftovers are eclipse hours (it assumes the same day shape every day). STL robust: the 4 injected dropout hours rank **#4–#7**; after them nothing exceeds **1.75 A**. The top 3 are the first and last eclipse days (the season switching on and off).	–
17	Fake disaster: Aug 2024 = 94%. Classic August effect **−1.10 → −3.00**. STL robust keeps **−1.20** and puts **−3.99** in the residual.	(quiz, advanced)
18	36 months = 3 rainy seasons = the minimum.	"How many cycles before you trust it?" (quiz)
19	Battery voltage dips in **March (−1.89 V)** and **September (−1.70 V)**; trend **49.76 → 49.55 V** (ageing).	"Why two dips, not one?" (two equinoxes)

Quiz coverage (original Week 3 quiz, topic "decomposing an operational time series")

Quiz question	Answer	Where it's taught
How does the observed series relate to its parts (additive)?	trend + seasonal + residual	Slide 5, task A4
Which period for daily NOC tickets?	7	Slides 9–10, tasks B1–B3
Swing grows with the level: which model?	Multiplicative	Slides 12–13, tasks B4–B5

Quiz question	Answer	Where it's taught
Where should an incident review look for unusual months?	The residual	Slide 15, task B6
Minimum full cycles before trusting a seasonal profile?	Three	Slide 18, task A2
One catastrophic August makes every August look bad: fix?	STL with robust=True	Slide 17, bonus

Answer key

Task	Answer	Common mistake → what to say
A1	`"MS"`	`"M"` → "M means month **end** (and is being renamed); our labels are month **starts**"
A2	`len(kam) // 12` → 3	`/` gives 3.0, which is also accepted
A3	`12`	`7` → "that's for daily data"
A4	`parts.trend + parts.seasonal + parts.resid` → gap 0	–
A5	`parts.seasonal.idxmin().month` → 8 (August)	Using `.min()` gives the value, not the date
B1	`7`	`12` or `365`
B2	`round(weekly.loc["Sunday"], 1)` → −2.9	Typing 2.9 without the minus sign
B3	`12`	Answering 7 (the good one)
B4	`"multiplicative"`	–
B5	`np.log(demo["multiplicative"])`	Logging the whole `demo` table
B6	`aug_parts.resid.abs().idxmax()` → 2025-08-18 09:00	Forgetting `.abs()`: the dropout is a **negative** residual, so `.idxmax()` alone finds the wrong hour
B7	`[3, 9]`	Giving month names; numbers are needed
🌶️	`robust=True` → August seasonal ≈ −1.20	Leaving it blank: plain STL still says ≈ −2.5

Troubleshooting

Problem	Fix
`x must have 2 complete cycles`	Too few rows for the period. Usually the wrong period (e.g. 365 on 2 years of data)

Problem	Fix
`does not handle missing values`	NaN inside the series. In this course the files are clean, so check they didn't create gaps themselves
The decomposition picture appears twice	They wrote `parts.plot()` as the last line. Use `fig = parts.plot()` then `plt.show()`
STL feels slow on hourly data	Normal: a few seconds for 2,880 rows
Someone loaded the battery file without `week="week-03"`	They get the Week 4 version, whose extra fault is in the capacity column. For `eod_voltage_v` the answers come out the same, but ask them to fix it: it matters in Week 3

Notes on the original material

- The original Week 3 code book **describes** decomposition (periods, components) but contains **no decomposition code**. Everything in today's notebook is new.
- The code book says the Ka-band trend is "about +0.05 points a year". Measured from the decomposition, the trend rises from 99.24% to 99.40% over 23 months (about +0.08 a year). The slides quote the measured numbers.
- The `battery_health_daily.csv` in `week-03/data` and `week-04/data` are **different files**. Today uses the Week 3 one (no injected faults).